The finite boundaries for alternating-group rank

Human Compression project: a component of the L044 recovery

The finite values in Theorem 1.1 of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, are attributed there to Fernandes–Leemans–Mixer, *Polytopes of high rank for the alternating groups* (2012). We recover them here from accepted classifications and three additional complete executions of an already acquired enumeration algorithm. We also identify the exact degree-twelve boundary and the independent lower-existence input for the principal assembly. The new finite work concerns only degrees seven, eight and ten.

Theorem 1. *For $3 \le n \le 14$, the maximum rank of a string C-marking generating A_n is as follows; a zero means that no such marking exists:*

n	3	4	5	6	7	8	9	10	11	12	13	14
maximum	0	0	3	0	0	0	4	5	6	5	6	6.

Lemma 2 (The new complete finite executions)**.** *The complete ordered class counts for A_{10} in ranks three, four and five are respectively* $177, 4, 8$*, and every other rank has zero classes. Neither A_7 nor A_8 has a string C-marking.*

Proof. Use the complete common proof of enumeration by stabilizers of initial strings, retained in the L071 alternating-group component. We execute its unchanged program `enumerate_even.g` with $G = A_n$, normalizing group S_n, and the following full limits:

n	$\lvert A_n\rvert$	$\lfloor \log_2 \lvert A_n\rvert \rfloor$	transitive recorded tuples	tuples generating A_n
7	2520	11	0	0
8	20160	14	76	0
10	1814400	20	245	189.

The rank limits follow from independence: every appended distinguished generator lies outside the subgroup generated by its predecessors, so each step at least doubles the order. No expected maximum or source count is used to terminate the enumeration.

For clarity, the retained completeness argument applies without any extra orbit restriction. The program forms all nonidentity involutions from the conjugacy classes of the literal permutation group A_n. After a fixed prefix, it considers every involution outside its generated subgroup commuting with every earlier mark except possibly the last. Representatives are taken under the subgroup of S_n fixing the entire prefix; conjugation therefore preserves all earlier marks and all extension conditions. The recursive end-intersection criterion is applied only to prefixes already having the intersection property. It is the acquired criterion for string C-markings, retained as an explicit mathematical input (McMullen–Schulte, Proposition 2E16).

The possible-transitivity pruning step is necessary: every later mark belongs to the allowed commuting pool, so if that pool and the current prefix generate an intransitive group, no extension can generate A_n. A transitive proper prefix is *not* discarded; it is recorded and extended. Thus these executions do not impose the all-omissions-intransitive hypothesis of the different L044 Table 2 census. A prefix generating all of A_n cannot be extended independently and is terminal. These rules exhaust all C-markings through the subgroup-order limit.

All three executions terminate normally with their complete exhaustion record and unchanged program bytes. A separate exact execution checks every one of the 321 recorded transitive tuples: each mark is an involution of A_n, the string commutations hold, the subgroup is transitive, and all 2^{2d} pairs of marked subsets satisfy the intersection property. Group equality with the actual ambient A_n, rather than order or a type comparison alone, selects the generating tuples. It gives the displayed totals and the A_{10} rank counts $177, 4, 8$. In degree eight all recorded groups are proper, including the sole rank-five group of order 576; none is mistaken for A_8.

For $n = 7, 8, 10$, simultaneous S_n conjugacy is the full automorphism action on A_n, so these are ordered abstract marking counts. Reversal is not included in the equivalence. This standard automorphism theorem is retained, although the nonexistence and maximum-rank assertions require only the completeness of the conjugacy enumeration. Ranks one and two cannot generate a nonabelian simple group: one involution generates a group of order at most two, and two involutions generate a dihedral group with a cyclic subgroup of index at most two. This completes all ranks. □

Proof of Theorem 1. The group A_3 has no nonidentity involution. All involutions of A_4 belong to its normal Klein four group, so they do not generate A_4. For A_6, the accepted degree-six classification in *Commuting involutions and six-point polyhedra* proves that no generating rank-three C-marking exists. The complete L047 degree-six census separately excludes rank four in A_6. That latter search treats connected strings, which is sufficient: a disconnected C-diagram would express A_6 as the direct product of the subgroups on its components. Their intersections are trivial by IP, they commute, and each component is nontrivial, contradicting simplicity. The retained Whiston bound for independent families in a proper subgroup of S_6 is four, so no higher rank is possible. Ranks one and two are excluded as in Lemma 2.

The accepted complete L071 benchmark censuses give exactly three ordered markings of rank three for A_5, and $73, 11$ ordered markings of ranks three and four for A_9. Their full subgroup-order limits are five and seventeen, respectively, and all other ranks vanish. The accepted L071 alternating-group census gives, in ranks three through six, the four rows

A_{11}	122	0	0	4
A_{12}	375	173	43	0
A_{13}	3021	202	50	20
A_{14}	8531	255	90	18.

It exhausts every rank through the respective subgroup-order limits $24, 27, 31, 35$, with every remaining row zero. Thus in particular it excludes *all* rank-six and rank-seven C-markings of A_{12}, with no compatibility restriction on their types or selected generators. These acquired classifications, together with Lemma 2, give every entry of the theorem. None of those already accepted executions is repeated here. □

The independent lower-existence input

The accepted whole principal recovery of L038, Fernandes–Leemans, *String C-group representations of alternating groups*, arXiv:1810.12450v2, proves that for every $n \geq 12$ and every $3 \leq d \leq \lfloor (n-1)/2 \rfloor$, A_n has a string C-marking of rank d. Its complete proof consists of the finite-end construction paper, the signed-end construction paper and the rank-interval assembly, together with their actual acquired contraction and end-intersection inputs. Three earlier maximal constructions remain explicitly attributed theorem inputs in that recovery. They assert existence with generation and IP; their historical description as maximal constructions does not import the L044 upper bound.

Consequently the L038 existence theorem may be used as the lower half of the L044 principal statement without circularity. The L044 proof still has to establish its large-degree upper bound.

The finite classification above supplies degree twelve as a complete base, or degrees twelve through fourteen if that longer base is convenient. A strong induction above those bases must state every smaller-degree invocation explicitly; the existence theorem does not discharge it.

Finite premises and attribution. The new required finite component consists of the unchanged acquired enumeration engine, the acquired full-IP routine, the new guarded execution checker and the complete outputs for degrees seven, eight and ten. GAP exact group operations remain software premises. The checker rejects optimized Python before creating files and clears `PYTHONOPTIMIZE` from child environments. Its default mode repeats the complete enumerations and full-subset checks; the retained-author mode acquires the already completed, separately pinned first runs. Source classifications and the stabilizer method retain their original attribution. The new work supplies the exact missing boundary executions and the noncircular assembly interfaces. No complete L044 recovery, open-conjecture solution, Lean verification or compression measurement is claimed here.